

Supplemental Information for:
Distinct horizontal transfer mechanisms for type I and type V
CRISPR-associated transposons

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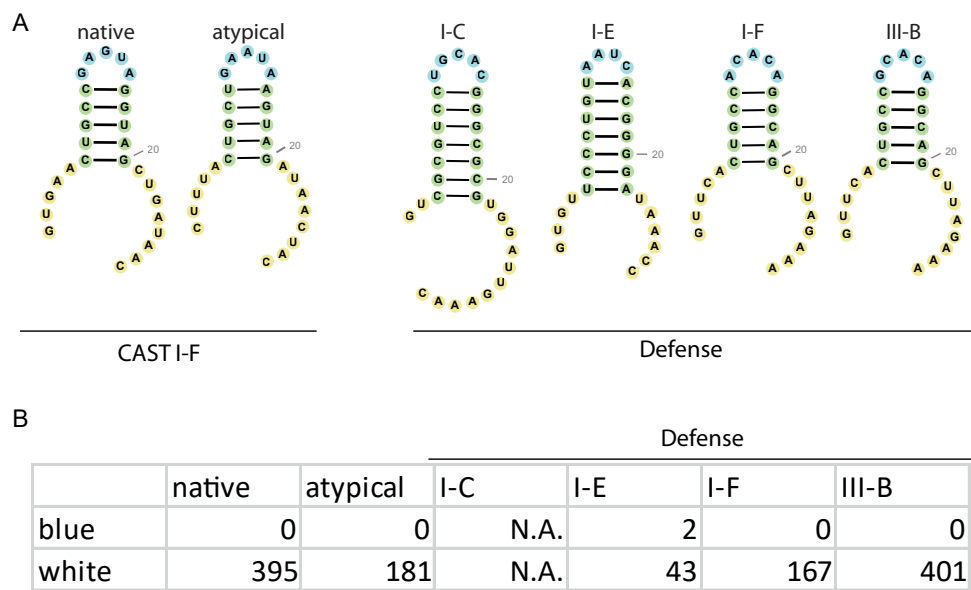


Figure S1: **A type I-F CAST catalyzes on-target transposition using heterologous CRISPR arrays.** (A) The structure of native type I-F CAST direct repeats (DRs) compared to (B) the structures of DRs from co-occurring CRISPR-Cas systems. Colors indicate different structural elements in the DR. Blue: loop; green: stem; yellow: handles. (C) On-target transposition into *lacZ* results in white colonies on LB+CmR+X-gal plates. In this example, the type I-F CAST and its native crRNA are guided to *lacZ* in the recipient cell genome. Integration is confirmed via junction PCR and Sanger sequencing. (D) On-target transposition remains at $\sim 100\%$ with DRs from the defense-associated type I-E, I-F, and III-B CRISPR arrays. The type I-C DR is not active (N.A.).

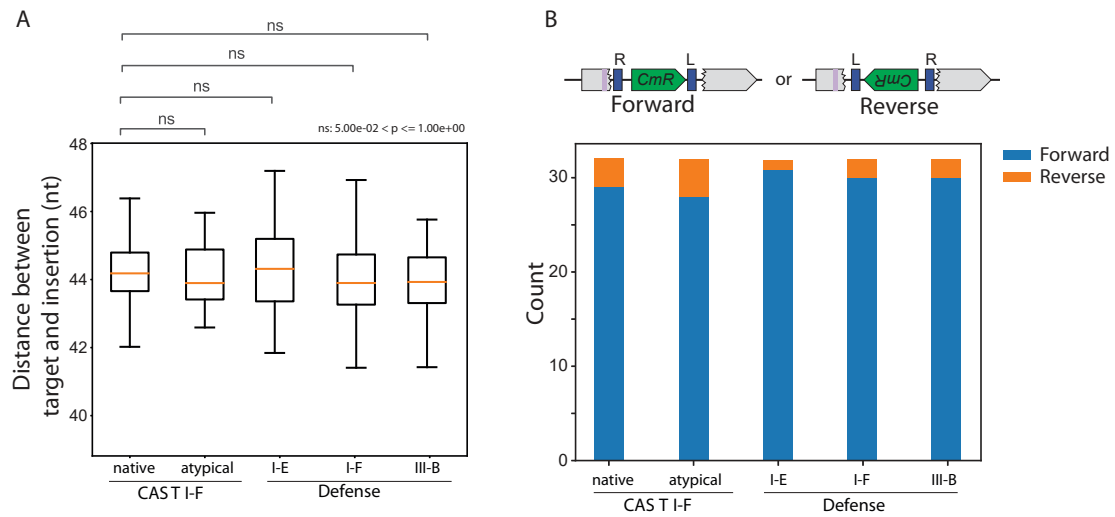


Figure S2: Defense-associated CRISPR RNAs support on-target integration. (A) The distance between the target site and the integration site for the indicated crRNAs, as determined via Sanger sequencing of 32 colonies for each condition. n.s.: $p > 0.05$. (B) Schematic of forward and reverse integration orientations (top). The integration orientation was determined via Sanger sequencing for each of the indicated crRNAs (bottom).

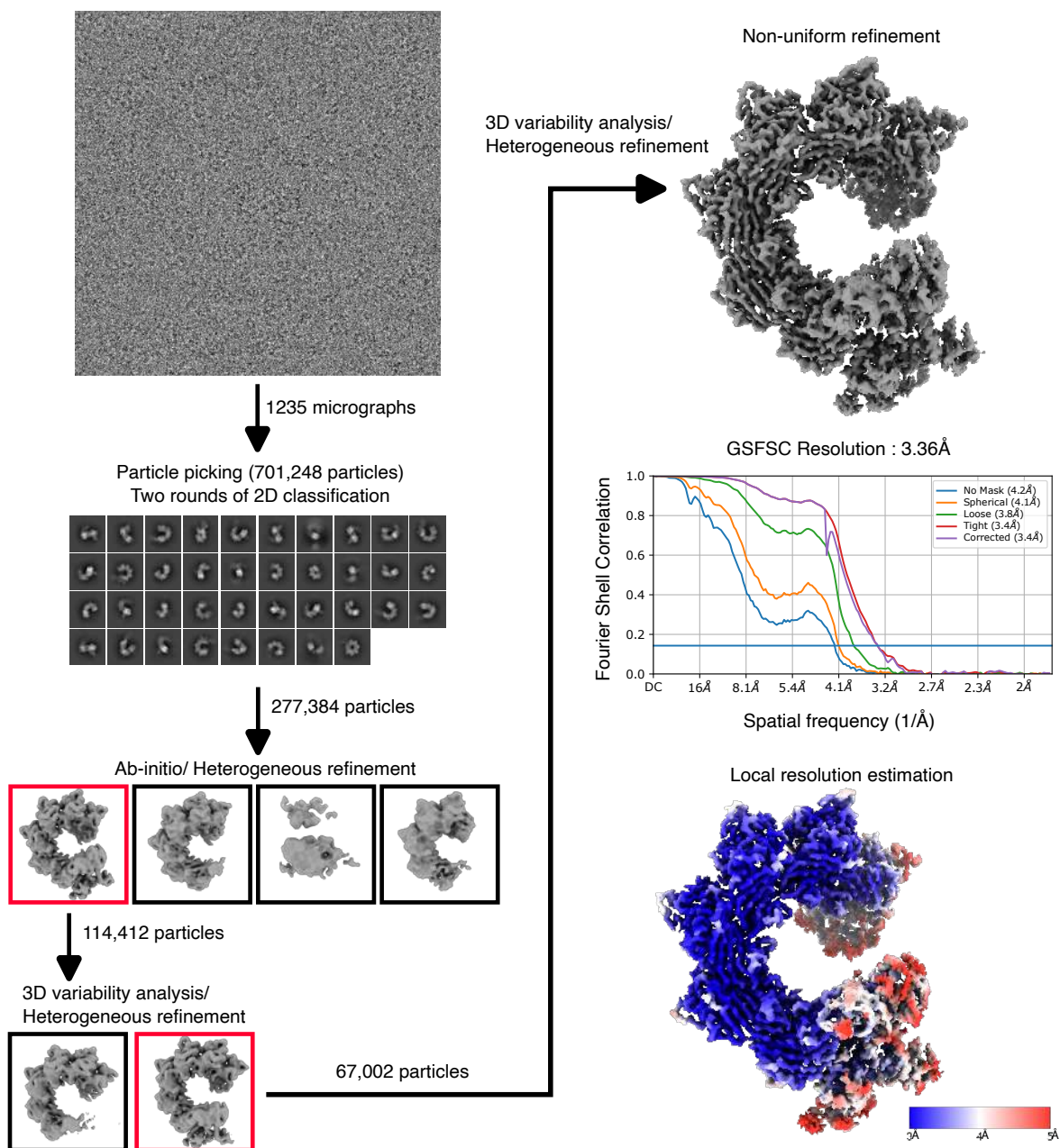


Figure S3: Cryo-EM image processing workflow for the TniQ-cascade with defense III-B array complex.

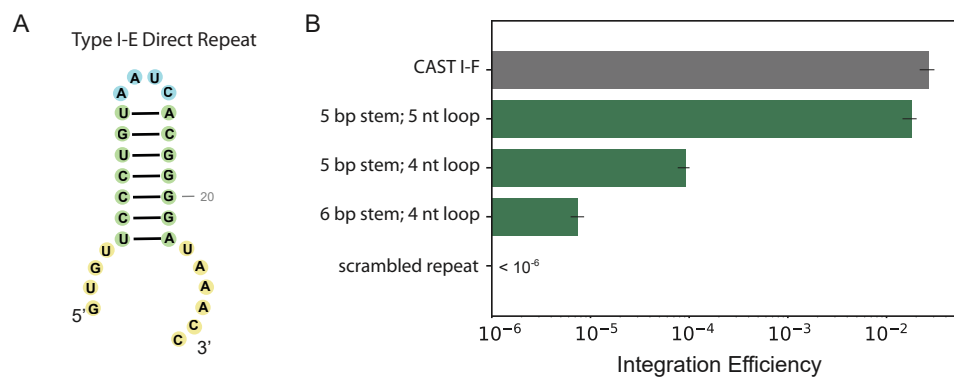


Figure S4: **The length of the direct repeat (DR) stem loop controls transposition.** (A) Predicted structure of the DR from a defense-associated type I-E CRISPR-Cas system. (B) Changing the loop and stem length increases integration efficiency. Error bars: mean of three replicates.

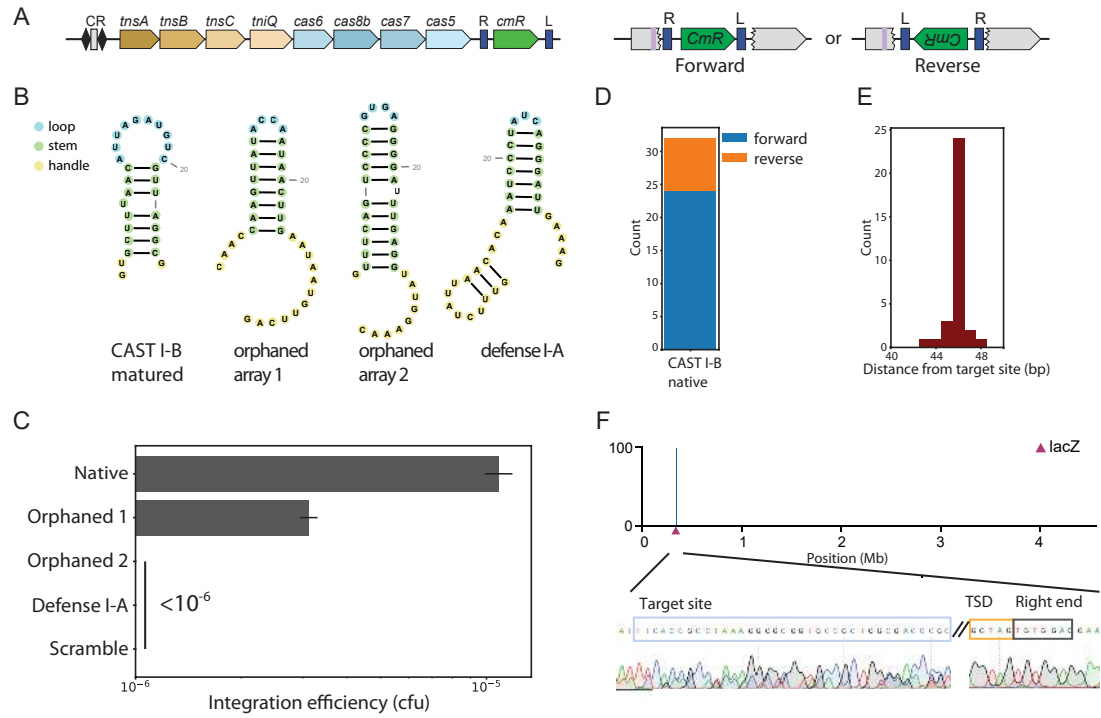


Figure S5: Type I-B CASTs co-opt spacers from heterologous CRISPR arrays. (A) Gene architecture of the type I-B *Av*CAST, cloned into the R6k plasmid (pIF1004, Table S1). (B) Predicted structure of the direct repeat (DR) from all CRISPR arrays found in *anabaena variabilis* (ATCC 29413). (C) Quantification of transposition from the native CAST array and co-occurring CRISPR systems. Error bars: standard deviation across three biological replicates. Scrambling the DR suppressed transposition below our detection limit of $\sim 10^6$ cfus. (D) The integration orientation and (E) distance between the target and insertion sites were determined via Sanger sequencing (N=32 clones). (F) Long-read NGS confirms single on-target cut-and-paste transposition into *lacZ*. Target site duplication (TSD) is also visible in this data.

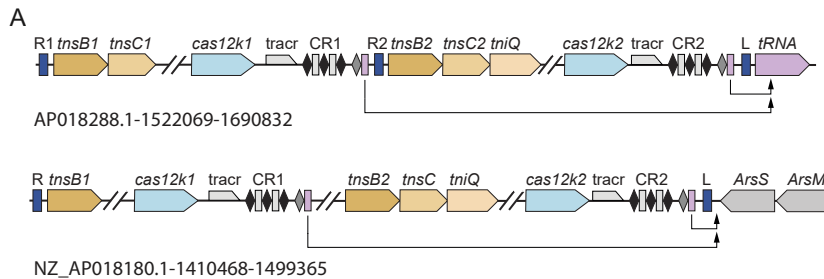


Figure S6: Examples of metagenomically-encoded type V CASTs that insert twice at the same genomic site. Assession numbers are indicated for each example.

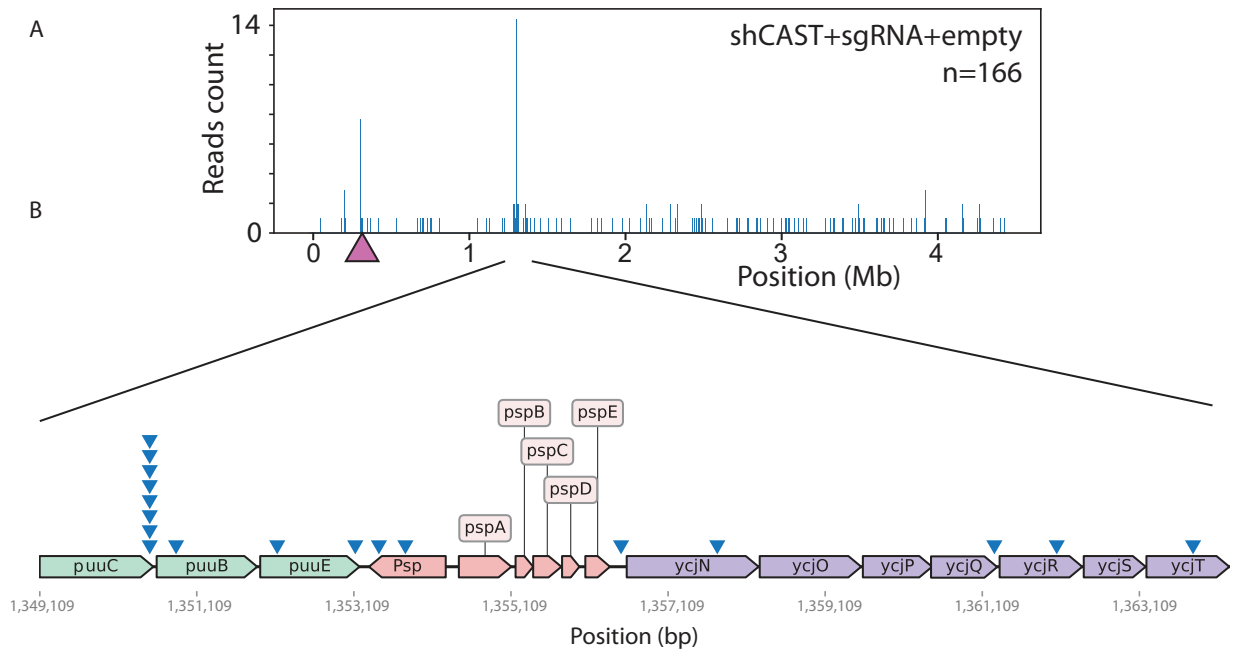


Figure S7: High resolution view of the off-target genomic hotspot during ShCAST-catalyzed transposition (A) In the absence of S15, whole-genome long-read sequencing reveals an apparent integration hotspot in the bacterial genome (N=166 insertion events). Diamond: on-target site (*lacZ*). (B) A higher-resolution view of the integration hot spot. Integration occurs over a broad region spanning multiple operons (green, pink, and purple). The events near *puuC* span a ~ 200 bp window without any strong sequence homology to the sgRNA. We conjecture that this hotspot may be due to a structure-specific feature of the bacterial genome.

Supplemental Tables

Plasmid	Description	Source
pIF1010	miniTn7-encoding R6K plasmid	Addgene: 64968 [1]
pIF1011	<i>V. cholerae</i> TniQ, Cas8, Cas7, and Cas6	Addgene: 130637 [2]
pIF1012	Kanamycin resistance cassette	Addgene: 130634 [2]
pIF1013	<i>V. cholerae</i> TnsA, TnsB, and TnsC	Addgene: 130633 [2]
pIF1014	ShCAST system and its native crRNA	Addgene: 127922 [3]
pIF1015	Kanamycin resistance cassette for ShCAST	Addgene: 127924 [3]
pIF1016	Expression of AvCAST proteins	Addgene: 168137 [4]
pIF1017	AvCAST donor	Addgene: 168145 [4]
pIF1008	R6K backbone with Golden Gate (GG) cloning sites	This paper
pIF1001	R6K plasmid encoding a type I-F CAST but no cargo	This paper
pIF1002	R6K plasmid with a I-F CAST targeting <i>lacZ</i>	This paper
pIF1003	R6K plasmid with a I-B CAST but no cargo	This paper
pIF1004	R6K plasmid with a I-B CAST targeting <i>lacZ</i>	This paper
pIF1005	R6K plasmid encoding a type V CAST but no cargo	This paper
pIF1006	R6K plasmid with a type V CAST targeting <i>lacZ</i>	This paper
pIF1008	Type I-F Cascade over-expression	This paper
pIF1009	Expression of a type III-B defense-associated crRNA	This paper

Table 1: Plasmids used in this study.

Data collection and Processing	
Microscope	Glacios
Pixel size (Å)	0.94
Voltage (kV)	200
Detector	Falcon IV
Exposure ($e^-/\text{Å}^2$)	40
Defocus range (μm)	-1.2 to -2.2
Final particles	66,862
Symmetry	C1
Map Resolution (Å)	3.36
(FSC threshold=0.143)	
Model refinement and validation	
Initial model used	6PIG
Ramachandran	
Flavor (%)	95.63
Allowed (%)	4.37
Outlier (%)	0.00
Rotamer outliers (%)	0.16
Clash score	7.88
MolProbity score	1.73
EM Ringer score	2.82

Table 2: Cryo-EM data collection and processing statistics

References

- [1] K.-H. Choi, H. P. Schweizer, mini-tn7 insertion in bacteria with single attTn7 sites: example *Pseudomonas aeruginosa*, Nature protocols 1 (2006) 153–161.
- [2] S. E. Klompe, P. L. H. Vo, T. S. Halpin-Healy, S. H. Sternberg, Transposon-encoded CRISPR–Cas systems direct RNA-guided DNA integration, Nature 571 (2019) 219–225.
- [3] J. Strecker, A. Ladha, Z. Gardner, J. L. Schmid-Burgk, K. S. Makarova, E. V. Koonin, F. Zhang, Rna-guided DNA insertion with CRISPR-associated transposases, Science 365 (2019) 48–53.
- [4] M. Saito, A. Ladha, J. Strecker, G. Faure, E. Neumann, H. Altae-Tran, R. K. Macrae, F. Zhang, Dual modes of CRISPR-associated transposon homing, Cell 184 (2021) 2441–2453.